

Lab 1

Query

Query History

```
2      SELECT
3          c.name AS category_name,
4          f.title,
5          f.length,
6          DENSE_RANK() OVER (
7              PARTITION BY c.name
8              ORDER BY f.length DESC
9          ) AS rank_num
10     FROM film f
11     JOIN film_category fc ON f.film_id = fc.film_id
12     JOIN category c ON fc.category_id = c.category_id
13 )
14 SELECT
15     category_name,
16     title,
17     length,
18     rank_num
19 FROM ranked_films
20 WHERE rank_num <= 3
21 ORDER BY category_name, rank_num;
```

Data Output Messages Notifications

Showing rows: 1 to 64 Page

	category_name character varying (25)	title character varying (255)	length smallint	rank_num bigint
5	Animation	Pond Seattle	185	1
6	Animation	Gangs Pride	185	1
7	Animation	Theory Mermaid	184	2
8	Animation	Sons Interview	184	2
9	Animation	Lawless Vision	181	3
10	Children	Wrong Behavior	178	1
11	Children	Fury Murder	178	1
12	Children	Empire Malkovich	177	2
13	Children	Model Fish	175	3
14	Classics	Conspiracy Spirit	184	1
15	Classics	Dracula Crystal	176	2
16	Classics	Lights Deer	174	3
17	Comedy	Control Anthem	185	1

Total rows: 64 Query complete 00:00:00.067

lab 2:

Query History

```
5      FROM payment
6      GROUP BY DATE_TRUNC('month', payment_date)
7  ),
8  growth_calc AS (
9      SELECT
10         payment_month,
11         current_revenue,
12         LAG(current_revenue) OVER (ORDER BY payment_month) AS previous_revenue
13     FROM monthly_revenue
14 )
15 SELECT
16     TO_CHAR(payment_month, 'YYYY-MM') AS month,
17     current_revenue,
18     previous_revenue,
19     ROUND(
20         ((current_revenue - previous_revenue) / previous_revenue) * 100,
21         2
22     ) AS mom_growth_rate_pct
23 FROM growth_calc
24 ORDER BY payment_month;
```

Data Output Messages Notifications

Showing rows: 1 to 4

	month text	current_revenue numeric	previous_revenue numeric	mom_growth_rate_pct numeric
1	2007-02	8351.84	[null]	[null]
2	2007-03	23886.56	8351.84	186.00
3	2007-04	28559.46	23886.56	19.56
4	2007-05	514.18	28559.46	-98.20

Lab 3

Query

Query History

```

1  SELECT
2      customer_id,
3      payment_id,
4      payment_date,
5      amount,
6      SUM(amount) OVER (
7          PARTITION BY customer_id
8          ORDER BY payment_date, payment_id
9          ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
10     ) AS customer_lifetime_value
11 FROM payment
12 ORDER BY customer_id, payment_date;

```

Data Output

Messages

Notifications

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SQL

Showing rows: 1 to 1000

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	customer_id smallint	payment_id [PK] integer	payment_date timestamp without time zone	amount numeric (5,2)	customer_lifetime_value numeric
1	1	18495	2007-02-14 23:22:38.996577	5.99	5.99
2	1	18496	2007-02-15 16:31:19.996577	0.99	6.98
3	1	18497	2007-02-15 19:37:12.996577	9.99	16.97
4	1	18498	2007-02-16 13:47:23.996577	4.99	21.96
5	1	18499	2007-02-18 07:10:14.996577	4.99	26.95
6	1	18500	2007-02-18 12:02:25.996577	0.99	27.94
7	1	18501	2007-02-21 04:53:11.996577	3.99	31.93
8	1	22680	2007-03-01 07:19:30.996577	4.99	36.92
9	1	22681	2007-03-02 14:05:18.996577	3.99	40.91
10	1	22682	2007-03-02 16:30:04.996577	0.99	41.90
11	1	22683	2007-03-17 11:06:20.996577	4.99	46.89
12	1	22684	2007-03-18 02:25:55.996577	0.99	47.88
13	1	22685	2007-03-19 08:23:42.996577	0.99	

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Successfully run. Total query

lab 4

```

8      RANK() OVER (
9          PARTITION BY co.country
10         ORDER BY SUM(p.amount) DESC
11     ) AS spend_rank
12 FROM customer c
13 JOIN address a ON c.address_id = a.address_id
14 JOIN city ci ON a.city_id = ci.city_id
15 JOIN country co ON ci.country_id = co.country_id
16 JOIN payment p ON c.customer_id = p.customer_id
17 GROUP BY co.country, c.customer_id, c.first_name, c.last_name
18 )
19 SELECT
20     country,
21     customer_id,
22     first_name,
23     last_name,
24     total_spend
25 FROM customer_spending
26 WHERE spend_rank = 1
27 ORDER BY country;

```

Data Output Messages Notifications

Showing rows: 1 to 108

	country character varying (50)	customer_id integer	first_name character varying (45)	last_name character varying (45)	total_spend numeric
1	Afghanistan	218	Vera	Mccoy	67.82
2	Algeria	176	June	Carroll	151.68
3	American Samoa	320	Anthony	Schwab	47.85
4	Angola	528	Claude	Herzog	93.80
5	Anguilla	381	Bobby	Boudreau	99.68
6	Argentina	560	Jordan	Archuleta	129.71
7	Armenia	41	Stephanie	Mitchell	118.75
8	Austria	266	Nora	Herrera	113.74
9	Azerbaijan	334	Raymond	Mcwhorter	108.75
10	Bahrain	590	Seth	Hannon	108.76
11	Bangladesh	21	Michelle	Clark	146.68
12	Belarus	144	Clara	Shaw	189.60
13	Bolivia	431	Joel	Francisco	

✓ Successfully run. To